

AI-driven gamified speech training for primary students: framework and evaluation

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ABSTRACT

Traditional public speaking education often suffers from limited learner engagement, delayed formative feedback, and a lack of interactive and adaptive training environments. This study proposes an AI-driven gamified speech learning framework (AI-GSLF), which combines real-time feedback technologies with motivational game design principles to address these issues. Based on this framework, a serious game—*Strongest Speech Streamer*—was developed using the Godot engine. The system integrates automatic speech recognition, sentiment analysis, and a novel speech rate detection algorithm to provide immediate feedback, helping learners adjust pacing, reduce anxiety, and enhance fluency during practice. A true experimental design was employed, involving 57 primary school students randomly assigned to either the experimental group using the gamified system or a control group following traditional methods over one month. Quantitative results showed that the experimental group demonstrated statistically significant improvements in motivation, confidence, speech accuracy, and delivery fluency. To our knowledge, few prior studies have integrated real-time AI feedback with systematic gamification for primary-level formal speech training. Findings support the potential of AI-GSLF as an effective, scalable approach to enhancing student performance and engagement in public speaking education.

1. Introduction

1.1. Importance of public speaking

The capacity for articulate and persuasive public speaking constitutes a fundamental skill in contemporary society [1–3], underpinning personal and professional efficacy across diverse contexts [4]. Early intervention at the primary school stage is particularly crucial, as this formative period plays a decisive role in building self-confidence, shaping communication habits, and reducing speech-related anxiety. In recognition of this, educational frameworks increasingly advocate for the cultivation of oral communication skills from the primary school level onward [5]. Such training is instrumental in developing students' logical thinking, mitigating social apprehension, and improving overall communication effectiveness [6]. Nevertheless, the widespread implementation of high-quality public speaking instruction is impeded by significant challenges, particularly the difficulty of providing timely, individualized feedback at scale. The present study seeks to address these limitations through the design of an AI-driven, gamified learning framework. This framework is engineered to enhance learner motivation

and skill acquisition through real-time, personalized guidance, offering a scalable and economically viable alternative to traditional methods.

1.2. Challenges in public speaking education

The systematic integration of public speaking into educational frameworks is, however, beset by formidable challenges. This is particularly evident in regions like East Asia, where pedagogical paradigms prioritize rote learning and written examinations over oral proficiency [7,8]. The impediments are multifaceted, encompassing the low curricular priority of oral communication, a paucity of specialized instructors, and the prohibitive costs of individualized coaching [2], and the absence of robust frameworks for assessment and real-time feedback [2]. This systemic de-emphasis is evident globally; while select institutions in China are adopting modern presentation formats and many Western schools offer speech courses [9], oral communication has yet to be institutionalized as a core competency in most educational systems [10,11].

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1.3. Opportunities with AI and gamification

The confluence of Artificial Intelligence (AI) and gamification presents a paradigm for surmounting longstanding pedagogical barriers in public speaking education. AI, through its capacity for speech recognition and natural language processing, offers automated, individualized feedback on critical delivery components like pronunciation and intonation [12], thereby facilitating deliberate practice without continuous instructor mediation [13]. The synergy with gamification, which leverages motivational affordances such as points and badges, cultivates intrinsic interest and persistence [14,15]. While the efficacy of gamification is well-documented in related domains like language learning [16], evidence suggests it can enhance knowledge acquisition and skill development through challenge-based tasks and instant feedback [15, 17]. However, despite such evidence, the current landscape of AI-enabled tools is characterized by a notable gap: a focus on adult users and general fluency (e.g., Speeko [12], Speakeasy) at the expense of structured pedagogy and age-appropriate design for formal youth education.

1.4. Purpose and research questions

To address these limitations, this study proposes an AI-driven gamified speech learning framework (AI-GSLF) that integrates real-time AI feedback with a theory-informed gamification structure, grounded in the Octalysis framework. Designed in alignment with Flow Theory, Deliberate Practice, and Cognitive Load Theory [18–21], the AI-GSLF aims to enhance practice engagement, learning motivation, and key public speaking competencies. The study specifically examines the following research questions:

- Q1: Can the AI-GSLF improve learners' practice frequency and behavioral engagement?
- Q2: Can the framework enhance learners' motivation and immersive experience?
- Q3: Can the AI-GSLF effectively develop learners' sub-skills in public speaking, including fluency, clarity, intonation, and content organization?

By bridging the gap between AI-driven feedback and gamified learning in structured speech education, this research aims to offer a scalable, cost-effective model that supports the development of public speaking skills among young learners, particularly in resource-constrained educational environments.

2. Literature review

2.1. Innovation in gamified assisted learning and speech learning

2.1.1. Background and advantages of gamified learning

The integration of multimedia technologies such as artificial intelligence (AI), virtual reality (VR), and augmented reality (AR) has significantly advanced the personalization and educational potential of computer games. Consequently, gamified learning—the use of game design principles in education—has emerged as a promising pedagogical model. By leveraging the interactive nature, feedback mechanisms, and challenge settings inherent in games, gamified learning aims to enhance learner motivation and engagement, thereby improving learning outcomes [22,23]. For instance, platforms like *Minecraft* have been used to teach programming and mathematics while fostering creativity [24], and applications like *Duolingo* improve language proficiency through task-driven approaches and progressive challenges [25]. These examples illustrate that well-designed gamified systems can increase learner interest and participation, leading to more effective learning and knowledge mastery [26].

In addition, gamified systems provide learners with clear learning

directions and real-time feedback, significantly enhancing motivation and learning outcomes [27,28]. They also create enjoyable learning experiences that strengthen motivation and optimize learning outcomes through well-designed game mechanics [29,30].

However, the efficacy of gamified learning is contingent upon the quality of its design [31,32]. As Huang and Hew [33] noted, if a game overemphasizes entertainment without effective alignment with learning objectives, learners may focus on gameplay at the expense of knowledge acquisition. Therefore, effective gamified learning requires structured designs in which game elements not only stimulate engagement but also guide learners toward intended outcomes. Research highlights three core mechanisms—rewards, challenges, and feedback—each of which plays a distinct role in shaping learner motivation and influencing educational performance [14,34].

2.1.2.1. Reward and challenge mechanisms. The design of reward mechanisms is crucial to enhancing learner motivation [35,36]. Common extrinsic rewards such as points, leaderboards, and achievements can increase short-term engagement and competition in game-based environments [32]. However, an over-reliance on these incentives may cause learners to focus on the rewards themselves rather than on knowledge acquisition, thereby diminishing long-term interest [36]. To sustain engagement, it is therefore critical to foster learners' intrinsic motivation, particularly their sense of autonomy, competence, and social relatedness [20]. Challenge mechanisms are a key approach for achieving this, as they provide clear goals and enhance a learner's sense of accomplishment through immediate feedback and dynamic difficulty adjustment [14,21,34].

Effective challenge design, however, must be carefully calibrated; challenges that are too difficult can lead to frustration, while those that are too easy may cause boredom [21]. Consequently, the design of challenge mechanisms should be guided by several key principles, including adaptive difficulty and clear goal-setting [34]. Adaptive difficulty, a core concept in Flow Theory, posits that learners are most engaged when a task's challenge level matches their current ability, inducing a "flow state" [21,37]. This can be implemented through systems that dynamically adjust task difficulty based on performance, as demonstrated in the educational game *RealGame*, where aligning challenges with player skill significantly increased immersion and engagement [38,39]. Clear goal-setting is another critical principle, as well-defined objectives increase motivation [14]. In a speech-learning context, this can be achieved with tiered tasks—such as progressing from a 30-second introduction to a 5-minute presentation—to scaffold learning and enhance a sense of growth [40]. As Kiili et al. [39] affirmed, the balance of challenge and skill, clear goals, and immediate feedback are essential design principles for creating an optimal learning experience.

2.1.2.2. Principles and optimization of feedback mechanisms. While immediate feedback is a critical component of gamified learning [38], its application to public speaking presents unique challenges. In fields like mathematics or programming, feedback often consists of simple success-rate analysis or error correction [32,41]. In contrast, public speaking is a multidimensional skill encompassing prosody, rhythm, and logical expression. Consequently, generic gamified feedback, such as points or level progression, is insufficient as it fails to provide the detailed, actionable guidance necessary for skill refinement. This limitation highlights a need to develop more sophisticated feedback mechanisms that can support learner self-regulation in this domain.

Traditionally, this detailed feedback is provided by instructors [42]. However, human feedback, while valuable, is often constrained by a lack of immediacy, potential inconsistency, and significant time and resource costs [2,12], particularly in large-scale educational settings. Recent studies suggest that AI-driven systems can overcome these limitations [43] by providing immediate, objective, and scalable speech

analysis that emulates an instructor's role.

This is supported by a comprehensive scoping review by J. J. Chen [44] identified AI's key affordances in education, including its function as an interactive companion and an adaptive personalization engine. Critically, the review also highlighted under-addressed challenges, such as the need for validated ethical guidelines and equitable access. These findings reinforce the necessity for frameworks like the proposed AI-GSLF, which strategically embeds AI-driven feedback within an inclusive and ethically informed pedagogical design.

2.1.3. The impact of gamified learning on immersion experience and learning motivation

Immersion and learning motivation are widely recognized as key factors that influence learners' long-term engagement and academic performance [45–47]. Research in immersive virtual reality (VR) environments, for instance, has demonstrated that a high degree of immersion enhances situational interest, reduces cognitive load, and improves knowledge acquisition [48]. These principles extend directly to gamified learning, which leverages interactive challenges and real-time feedback to create immersive experiences that foster sustained motivation [15].

Recent advances in educational technology further illustrate this potential. For example, Dæhlen et al. [49] developed a serious game with integrated eye-tracking for children with neurodevelopmental disorders, demonstrating that real-time physiological data can be used to dynamically adapt challenges and personalize feedback, thereby enhancing engagement [49]. Similarly, a study by Hwang et al. on a two-tier test-based gaming approach found that it significantly increased students' interest and active participation, leading to improved academic performance [50]. These studies confirm the effectiveness of well-designed interactive systems in promoting learning motivation.

To empirically assess these constructs, standardized instruments are commonly employed in the literature. The Immersion Experience Questionnaire (IEQ) is widely used to measure key dimensions of immersion, such as cognitive involvement and perceived control [51]. Concurrently, the Motivated Strategies for Learning Questionnaire (MSLQ) is a standard tool for assessing motivational components like self-efficacy and intrinsic motivation [52]. This study adopts these validated instruments to evaluate the impact of the AI-GSLF, ensuring a rigorous and objective assessment of its effects on immersion and motivation.

2.2. Limitations of traditional public speaking instruction and attempts at improvement

2.2.1. Limitations of traditional public speaking instruction

While gamified learning has shown potential in other fields, public speaking education is predominantly delivered through a traditional instructional model [53]. This approach typically involves a sequence of teacher lectures, student practice, in-class presentations, and subsequent instructor feedback [53,54].

Although widely adopted, this model is characterized by several significant limitations. A primary challenge is the high level of performance anxiety students often experience, which adversely affects their delivery [55]. Indeed, research indicates that nervousness and the lack of effective feedback are principal reasons for students' negative attitudes toward classroom presentations [56]. Furthermore, opportunities for practice are often scarce; in many classroom settings, students may only deliver a single formal speech, which is insufficient for targeted training and continuous skill refinement [53]. Finally, the feedback mechanism itself is frequently constrained. Feedback is often delayed, impression-based, and delivered long after the performance, by which point students may be unable to accurately recall their delivery, thus hindering effective improvement [55].

2.2.2. The "Lightning talk" improvement method and its theoretical foundation

As a non-technological attempt to address the challenges of public speaking traditional instruction, Jean-Pierre, Hassan, and Sturge [57] proposed the "Lightning Talk" method. This approach involves short, frequent public speaking exercises to reduce student anxiety. The efficacy of this approach can be understood through the theoretical lenses of Deliberate Practice and Cognitive Load Theory. Deliberate Practice Theory posits that skill acquisition requires repeated, goal-oriented practice with timely feedback, which aligns with the "Lightning Talk's" emphasis on frequent trials and immediate self-reflection [18]. Concurrently, Cognitive Load Theory suggests that breaking down complex tasks into smaller, shorter segments reduces the load on working memory, thereby improving learning efficiency [19]. By offering more practice opportunities in a structured format, this method allows students to improve fluency and adjust their strategies in a targeted manner [57]. Importantly, this insight inspired the design choice in this study to adopt **one-minute speech tasks** in the AI-GSLF, ensuring that learners could engage in intensive practice within a limited time frame and through small task segments. While the Lightning Talk effectively enhances public speaking skills and mitigates some shortcomings of the traditional model, it still relies on conventional classroom settings and is therefore constrained by limitations in class time, teaching resources, and scalable practice opportunities. By embedding the principles of "Lightning Talk" into an AI-driven, gamified platform, the current study aims to amplify its advantages—namely low cognitive load, reduced anxiety, and high efficiency in self-regulation—while overcoming the reliance on classroom-based supporting resources inherent in traditional implementations.

2.3. AI-Driven feedback in language and speech learning

2.3.1. The role of feedback and the rise of AI technologies

While recent advances in artificial intelligence have led to the development of speech training systems capable of automated scoring and feedback [58,59], these tools often lack the motivational mechanisms necessary for sustained learner engagement [12]. Conversely, gamified learning excels at boosting participation through challenges, rewards, and instant feedback [15].

The synergy of these two approaches—integrating AI-driven, real-time feedback with a robust gamification framework—therefore presents a promising solution for enhancing both skill development and learner motivation in speech education. However, the effective integration of these technologies requires more than technical accuracy. As scholars caution, without addressing critical concerns such as privacy, bias, and potential interpersonal disconnection, even the most advanced AI systems may fail to foster long-term learning outcomes [60]. Acknowledging this, the present study introduces the AI-driven Gamified Speech Learning Framework (AI-GSLF) to address these gaps by embedding technically accurate AI feedback within a motivational, engaging, and ethically considerate design.

2.3.2. The AI feedback mechanism as a substitute for teacher guidance in speech training

Effective feedback is widely recognized as a critical component of public speaking instruction, as it significantly improves performance in areas such as vocal modulation and speech organization [42]. However, research by Smith and King highlights that the efficacy of feedback is contingent upon factors like delivery intensity and learner sensitivity [42]. For instance, low-intensity feedback (gentle suggestions) is often more effective for highly sensitive students, whereas high-intensity feedback (direct criticism) can provoke defensive reactions [42]. This finding underscores the need for personalized feedback, a challenge that can be addressed by recent advances in AI.

AI-driven systems offer a promising solution by providing immediate, objective, and personalized analysis of speech metrics like

pronunciation, fluency, and speech rate [12]. Studies by Zou et al. [43] and Tai and Chen [61] demonstrate that such systems help learners understand their performance and receive targeted recommendations, effectively emulating the role of an instructor without the interpersonal complexities of human evaluation. Yet, while AI can optimize the feedback mechanism, it does not inherently address the broader challenge of sustained learner motivation. Effective gamification depends on a more comprehensive set of motivational drivers beyond real-time feedback alone [62]. Therefore, this study integrates AI feedback within the Octalysis framework to construct a holistic system designed to enhance not only skill acquisition but also long-term engagement and immersion.

2.4. Theoretical frameworks

2.4.1. Applying the octalysis gamification framework in learning systems

The Octalysis gamification framework is increasingly being applied across diverse educational domains [63]. Its effectiveness has been demonstrated in fields such as language learning [16], online platform evaluation [64], and programming education [65]. Moreover, studies on role-playing systems for corporate, medical, and military training have validated their capacity to enhance learner immersion and skill mastery [66]. Research consistently shows that applications designed using this framework can significantly improve user motivation, engagement, and immersive experience [67,68]. Despite this broad utility, its application to public speaking education remains unexplored. To date, there is a notable gap in the literature regarding how Octalysis-based gamification can be systematically leveraged to improve speech performance, reduce public speaking anxiety, and optimize personalized learning pathways.

2.4.2. Overview of the octalysis gamification framework

The Octalysis framework, proposed by Chou [62], is a model grounded in motivational psychology designed to analyze and build gamified systems that drive behavioral change. The framework identifies eight core motivational drivers: (1) Epic Meaning & Calling, which instills a sense of purpose; (2) Development & Accomplishment, which appeals to the intrinsic desire for progress and skill mastery; (3) Empowerment of Creativity & Feedback, which encourages strategic thinking and exploration; (4) Ownership & Possession, which fosters attachment and control over virtual items; (5) Social Influence & Relatedness, which leverages competition and collaboration; (6) Scarcity & Impatience, which creates urgency through limitations; (7) Unpredictability & Curiosity, which maintains engagement through surprise; and (8) Loss & Avoidance, which motivates action to prevent negative outcomes [62].

The framework further categorizes these drivers as either extrinsic (e.g., accomplishment, ownership, scarcity) or intrinsic (e.g., creativity, social influence, unpredictability). It also distinguishes between "White Hat" gamification, which uses positive motivators like purpose and mastery to create feelings of control and satisfaction, and "Black Hat" gamification, which uses drivers like scarcity and loss avoidance to create urgency [62]. The adaptability of this framework makes it particularly suitable for young learners. For this demographic, drivers such as epic meaning, immediate feedback, and social influence can enhance engagement and motivation, while loss-based mechanics can be implemented carefully to avoid undue stress. This targeted, age-appropriate approach makes Octalysis a practical blueprint for designing the motivational architecture of the AI-GSLF [62].

2.5. Summary of gaps and rationale for the study

Although gamified learning has shown potential across many fields, its application in public speaking education remains underexplored. Traditional instruction in this domain is often constrained by high student anxiety, limited practice opportunities, and a lack of immediate feedback. While pedagogical innovations like "Lightning Talks" address

some of these issues by applying principles from Deliberate Practice and Cognitive Load Theory, they are still limited by the physical classroom setting. Concurrently, AI offers a powerful solution for providing real-time, objective feedback, yet existing AI-based training systems often lack the motivational architecture required for sustained learner engagement.

A significant gap, therefore, exists at the intersection of these domains. While theoretical models like Flow Theory and the Octalysis framework provide a blueprint for designing motivating learning experiences, no comprehensive system has yet integrated AI-driven feedback with a robust, theory-informed gamification structure for public speaking training. This study aims to address this gap by proposing and evaluating the AI-driven Gamified Speech Learning Framework (AI-GSLF). The framework is designed to synergize the immediate, personalized feedback of AI with the engaging, motivation-sustaining mechanics of gamification to create an effective and scalable solution for public speaking instruction.

3. Research methodology

3.1. Construction of the ai-gslf

This study introduces the AI-driven Gamified Speech Learning Framework (AI-GSLF), a multi-component system designed to integrate motivational design, pedagogical structure, and technological support. The framework comprises three core modules. First, the **Gamification Module**, grounded in the Octalysis framework [62] and Flow Theory, is engineered to enhance user engagement and motivation. Second, the **Teaching Module** provides a systematic learning path based on principles from Deliberate Practice, Cognitive Load Theory, and the Feynman Technique [18,19,69]. Third, the **Technology Module** incorporates AI-powered, real-time feedback using speech recognition, feature analysis, and a Large Language Model (LLM) to provide expert-level performance evaluation.

The AI-GSLF was developed to address the distinct limitations of existing approaches. Systems that rely solely on AI, while providing precise feedback, often fail to sustain learner engagement. Conversely, purely gamified systems can maintain interest but typically offer superficial feedback (e.g., points and badges) that is insufficient for complex skill development in public speaking. Our integrated framework overcomes these challenges by synergizing technology and pedagogy. The AI-driven feedback ensures that each practice session is a targeted instance of "deliberate practice," while the gamification architecture fosters the intrinsic motivation required for sustained effort. Tailored for young learners, this dual focus on structured guidance and sustained engagement creates a more effective and supportive educational experience. To operationalize this framework, we developed a serious game, *Strongest Speech Streamer*, which serves as the practical embodiment of the AI-GSLF, as illustrated in Fig. 1.

3.2. AI-GSLF gamification module: design of the strongest speech streamer

3.2.1. Introduction to the strongest speech streamer

Based on the AI-GSLF, this study designed and developed a game titled *Strongest Speech Streamer*, which was implemented using the Godot engine and integrated with AI technologies to provide real-time feedback. The game supports both PC and Android platforms. In the game: (1) **Role Design**: Players take on the role of a newly recruited live streamer in a growing broadcasting company, using their work to improve speech skills. (2) **Gamification Elements**: The game incorporates various Octalysis-inspired systems such as: "Role-playing", "Leaderboards", "Shop system", "Rewards", "Punishments", "Achievements" and "Item system" (A detailed summary table mapping AI-GSLF game functions to their corresponding Octalysis gamification drivers is provided in Appendix 3 Table App. 2 for reference). (3) **Immersion**: A first-person perspective simulates a real speech stage, enhancing

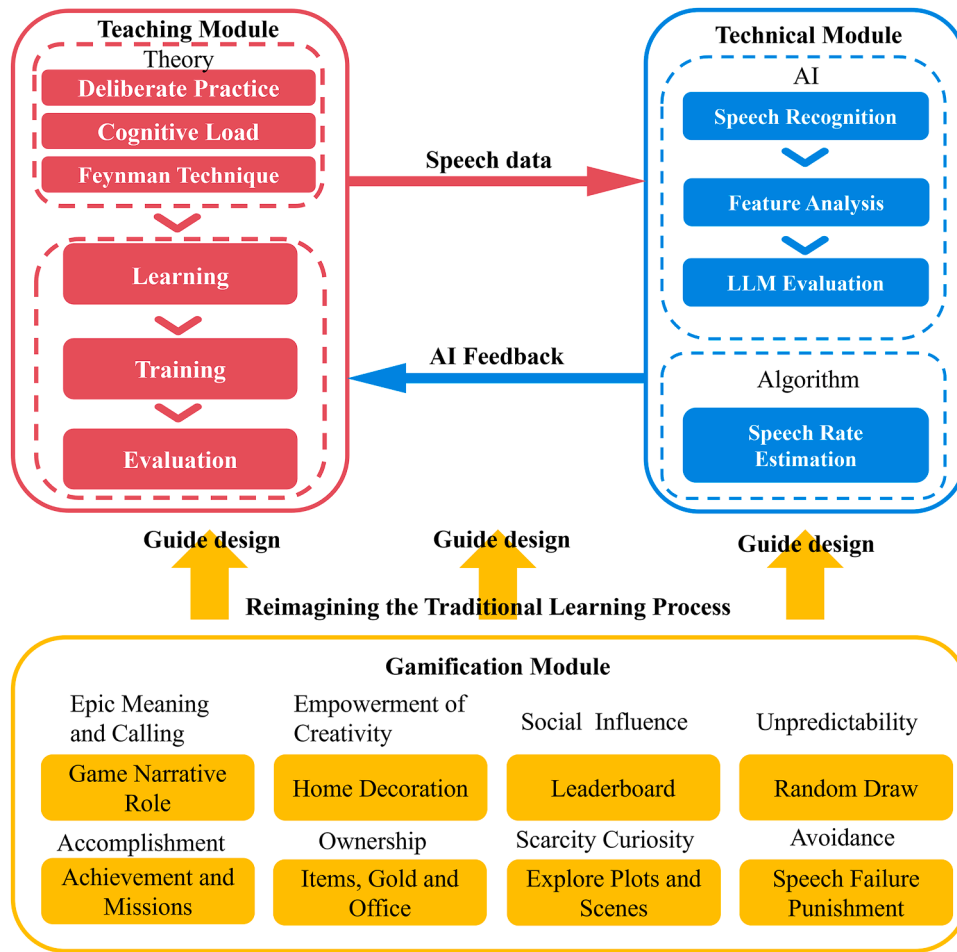


Fig. 1. AI-driven gamified speech learning framework.

presence during training. (4) **Speech Scenarios:** The game features real-world scenarios like workplace debriefing, campus speeches, and debates to build practical speaking skills. To ensure reproducibility, the source code and executable version of Strongest Speech Streamer are provided as supplementary materials (see Appendix 1 Supplementary Game Repository 1 for details).

3.2.2. Main gameplay flow

The gameplay is structured around a core instructional loop, where each in-game "day" represents a complete learning cycle. This cycle consists of four stages: theoretical learning, task preparation, speech delivery, and feedback. Players begin by studying speech theory via the in-game *Streamer Handbook*. They then proceed to their *Workstation* to receive a system-assigned task, after which they enter the *Broadcast Room* to deliver a one-minute impromptu speech. Upon completion, the integrated AI system provides structured, personalized feedback, which is archived in the player's *History* for review. To introduce a strategic element, players must manage a limited resource—stamina points—to balance practice and theoretical study. The ultimate objective is to maximize skill development and experience points to meet specified performance goals within a set timeframe.

3.2.3. Gamification mechanism design

The gamification design of *Strongest Speech Streamer* is grounded in the theory of learning motivation proposed by Schunk [45] and the eight core motivational drivers of the Octalysis framework [16]. To foster sustained engagement, the system integrates a dual-mechanism approach of reward and punishment.

The reward mechanism is designed to stimulate intrinsic motivation

and provide a sense of accomplishment. Players earn experience points and coins based on their performance, which facilitates leveling up, title progression, and company growth. Coins can be used to purchase virtual items, including consumables that improve speech scores and collectibles that unlock achievements and decorative objects. A lottery system, which leverages the Octalysis drives of Unpredictability & Curiosity and Scarcity & Impatience, provides random opportunities to acquire rare items, further enhancing motivation [16]. Conversely, a punishment mechanism, based on the Loss & Avoidance drive, imposes penalties such as wage or score deductions for poor performance, discouraging complacency and motivating improvement [16].

To further enhance engagement, the game incorporates several other mechanics. A character costume and home decoration system, grounded in the Empowerment of Creativity & Feedback drive, allows for personalization [16]. Drawing on research showing that formal attire can enhance social influence, players can acquire clothing to improve their professionalism score during speeches [70]. The home system allows players to decorate their space with collected items, some of which unlock additional training opportunities. Social features, including leaderboards and achievement showcases, leverage the drives of Ownership & Possession and Social Influence & Relatedness to foster competition and a sense of community [71]. Finally, hidden achievements tap into Unpredictability & Curiosity to encourage exploration [16]. Collectively, these design elements aim to maximize the time and effort learners invest in the game. The main game interfaces are illustrated in Fig. 2, with additional scenes available in Appendix 2.

Having detailed the motivational architecture of *Strongest Speech Streamer*, we now turn to the system's pedagogical core: the speech training module, which leverages proven educational theories to



Fig. 2. Main interfaces of *Strongest Speech Streamer*.

systematically enhance public speaking skills.

3.3. AI-GSLF speech training module: instructional design guided by learning theories

The speech training module in *Strongest Speech Streamer* is structured around a pedagogical approach that prioritizes theoretical understanding before practical application. This design is grounded in established educational psychology principles, including Deliberate Practice Theory, Cognitive Load Theory, and the Feynman Technique, with the objective of systematically enhancing players' public speaking skills.

The user interface for the speech training process is illustrated in the four panels of Fig. 4. Specifically, the top-left panel presents learning materials, the top-right panel displays assigned speech tasks, the bottom-left panel provides AI-generated feedback and suggestions, and the bottom-right panel shows the overall speech performance score. Furthermore, to enhance player immersion, the module incorporates a variety of simulated speech scenarios designed to mirror diverse, real-world situations, examples of which are depicted in Fig. 5.

3.3.1. Speech theory learning

Prior to practical training, players engage in a theoretical learning stage designed to establish a foundational knowledge framework for public speaking. This stage covers essential skills, including structured expression, emotional regulation, and non-verbal communication. The content is presented in modular, micro-unit lessons to facilitate a clear and manageable learning process.

This instructional design is informed by Cognitive Load Theory [19], which suggests that segmenting complex information into smaller components reduces extraneous cognitive load for novice learners. Moreover, this initial theoretical instruction serves as a critical foundation for subsequent practical application. This approach aligns with the principles of Deliberate Practice Theory [18], which posits that effective practice is contingent upon a clear understanding of the target skill. Consequently, this structure ensures that players can consciously and effectively refine their abilities during the subsequent training activities.

3.3.2. Speech practice training

Following the acquisition of foundational theoretical knowledge, players proceed to the speech practice phase. This phase comprises two distinct training components: speech delivery and technical skills. The design emphasizes frequent, targeted practice to foster the gradual development of comprehensive public speaking competence.

Speech delivery training involves one-minute practice sessions designed to simulate impromptu speaking scenarios. This format requires players to organize and articulate their ideas effectively within a constrained timeframe. This approach is informed by research on the efficacy of impromptu speaking, which indicates that short, intensive practice sessions can improve non-verbal communication skills, including eye contact, gestures, and vocal control [72]. Furthermore, this method aligns with Deliberate Practice Theory by incorporating specific goals, repeated practice, and immediate feedback to facilitate incremental skill refinement [18]. The one-minute task serves as a high-frequency, focused training exercise intended to reinforce skills and improve performance under pressure.

The speech topics are primarily derived from primary and secondary school curricula (e.g., Chinese, mathematics, English), an approach based on the Feynman Technique [69]. Applying this technique to public speaking training is intended to develop several core competencies concurrently. First, it promotes a shift from superficial memorization to deep comprehension, as explaining a concept requires a thorough understanding of its underlying principles. This internalization of knowledge can reduce speech anxiety associated with memory recall. Second, the act of teaching a topic necessitates logical structuring, compelling the speaker to organize thoughts into a coherent narrative. Third, the technique cultivates clarity and audience-centered communication by requiring the speaker to translate complex ideas into accessible language. This process improves verbal precision and enhances the ability to connect with an audience, thereby increasing both cognitive organization and communication efficiency in pressure situations.

To further optimize training effectiveness, the module incorporates several established learning theories. In accordance with Cognitive Load Theory [19], content is structured to prevent cognitive overload. The

design also draws on Flow Theory [21], which posits that optimal engagement occurs when task difficulty aligns with an individual's skill level. To achieve this balance, an Adaptive Difficulty Mechanism is implemented. This system dynamically adjusts task difficulty based on player performance, increasing the challenge as skills improve and reducing it to mitigate frustration when performance declines, thereby sustaining motivation [73]. This dynamic adjustment enhances the specificity of the training and aligns with the "optimal challenge" principle of deliberate practice [18]. A simplified flowchart of this mechanism is presented in Fig. 3, and the corresponding pseudocode is available in Appendix 2 (Supplementary Code 1).

3.3.3. Speech technique training

Targeted training for speech techniques is provided to enhance players' vocal expression, including volume control, intonation, and clarity of speech [72]. To develop these specific skills, *Strongest Speech Streamer* features three focused mini-games. The "Go to the Park" mini-game trains volume control by requiring players to maintain their voice within a specified range. "Catch the Bus" addresses speech rate, guiding players to keep their speaking speed within an optimal pace. Finally, "Play Arcade Games" focuses on rhythm, where players use specific rhythmic cues to guide their in-game character.

In summary, the speech training module integrates Deliberate Practice Theory, Cognitive Load Theory, and the Feynman Technique into a structured, multi-stage training approach. This design aims to systematically improve players' verbal organization, expressive delivery, and overall public speaking competency. Having outlined the pedagogical foundations of the module, the following section describes the technical architecture that supports this instructional design. Specifically, it will detail the AI-driven mechanisms used to provide real-time feedback and personalized speech assessments.

3.4. AI-GSLF technical module: technological realization

3.4.1. Real-time feedback mechanism based on AI technology

As established in the literature review (Section 2), feedback is a critical component of both gamified learning and speech training. However, traditional teacher-led feedback, while professional and targeted, is often limited by challenges related to immediacy, consistency, and scalability. The advancement of artificial intelligence (AI) offers potential solutions to overcome these limitations.

In this study, an AI-driven feedback module was developed for the AI-GSLF, designed to provide immediate and personalized evaluations. The module is built upon three core technologies: speech recognition, acoustic feature analysis, and large language models (LLMs). For cost-

effective and stable performance, the system utilizes Baidu Cloud's Qianfan AppBuilder platform for speech recognition and feature extraction. The Qianfan-Agent-Speed-8 K large language model serves as the core engine for semantic analysis and feedback generation. The selected model was a pre-trained foundation model and was not fine-tuned on participant data for this study; instead, its functionality was guided through prompt engineering tailored to the specific evaluation tasks. Key parameters included diversity (temperature), maximum output tokens, and the number of referenced dialogue turns, which were adjusted within practical ranges to balance response quality and computational efficiency (see Appendix 3 Table App. 1 for details).

The prompts were designed to instruct the LLM to function as an expert speech evaluator. The key instructions specified that the model should: (1) objectively evaluate speech content and verbal expression, assigning a score from 0 to 100; (2) provide constructive feedback on content and delivery, including specific suggestions regarding volume, pace, and rhythm; (3) generate feedback that is professional in tone and adheres to a consistent format, comprising a total score and actionable suggestions for improvement; and (4) use a five-level scoring rubric (in 20-point increments) to ensure standardized evaluation. These components were integrated into an application, termed the "AI Speech Judge Expert," which was connected to the *Strongest Speech Streamer* game via an API.

The system was designed to provide personalized feedback by analyzing the unique audio data from each player's performance in real time. Instead of relying on pre-determined responses, the AI Speech Judge Expert processes the specific acoustic features (e.g., speech rate, volume, fluency) and semantic content of every submitted recording. This analysis results in diagnostic feedback and improvement suggestions tailored to the individual's performance. The high granularity and immediacy of this feedback ensure its direct relevance to the player's specific actions, thereby supporting a personalized learning trajectory. Example inputs, outputs, and a screenshot of the AI module's backend are provided in Appendix 2 (Fig. App. 4).

During development, a key technical requirement was identified: the speech recognition API only supported mono-channel audio input. To address this, the FFmpeg multimedia framework was integrated into the game to automatically convert stereo audio recordings to a compatible mono format, ensuring seamless interaction with the AI evaluation system. Furthermore, for efficient data management, Alibaba Cloud's Object Storage Service (OSS) was used to store all user-generated audio data and the corresponding AI analysis results. This centralized storage solution also facilitated data access for subsequent analysis.

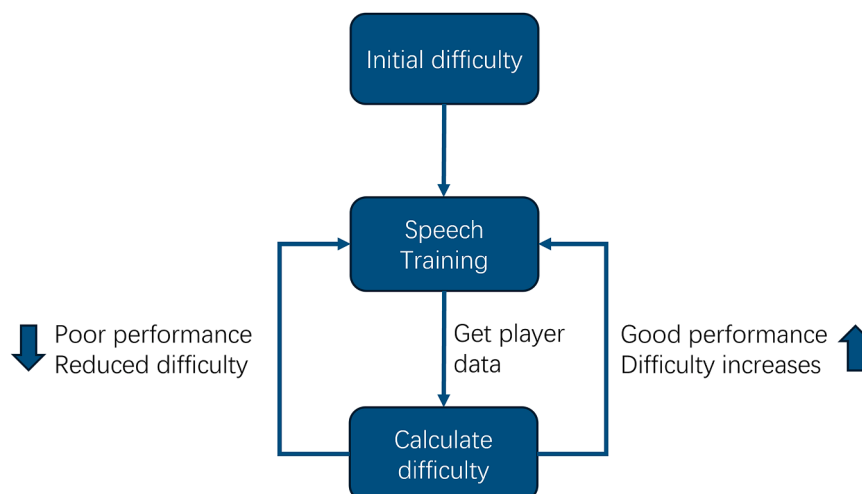


Fig. 3. Flowchart of the adaptive difficulty mechanism.



Fig. 4. Main interfaces of speech training.



Fig. 5. Part of the speech scenes.

3.4.2. Technical realization of speech training levels

The implementation of the speech training levels is based on interaction mechanisms that process real-time vocal input. During gameplay, signal processing techniques and audio feature extraction algorithms are employed to dynamically analyze the player's speech data. These signal processing steps include short-time Fourier transform (STFT)-based analysis for extracting frame-level vocal features [79]. A detailed description of these algorithms and the corresponding source code is

provided in Appendix 2.

These three training levels—volume, speech rate, and rhythm control—are designed to be complementary, forming a systematic approach to developing vocal delivery skills. The objective is to help players achieve finer control over these vocal parameters, thereby enhancing their overall expressiveness. The user interfaces for these levels are depicted in Fig. 6: the left panel shows the volume control level, the middle panel presents the speech rate training, and the right panel



Fig. 6. Three training levels for speech skills.

illustrates the rhythm control level. A video demonstration of the game’s core functionalities is available as supplementary material (see Supplementary Video 1).

4. Experimental design

4.1. Experimental design and method

This study employed a true experimental design to evaluate the proposed learning framework. Participants were elementary school students recruited from a primary school in Kunming, Yunnan Province, China. This setting was selected for its cultural and educational relevance. In China’s examination-oriented educational system, oral communication skills are often underemphasized, resulting in limited opportunities for systematic speaking practice, suggesting that interventions incorporating AI and gamification may be particularly beneficial [7]. Furthermore, as existing research on such interventions has predominantly focused on Western and adult learner populations, this study addresses a significant geographic and demographic gap in the literature.

All experimental procedures received institutional approval, and informed consent was obtained from the school and the legal guardians of all participants. Initially, 63 students volunteered and were screened based on their baseline public speaking ability. These participants were then randomly assigned to either an experimental group or a control group using a computer-generated random number sequence. This randomized controlled trial design was chosen to enhance internal validity by minimizing selection bias and confounding variables, thereby strengthening the basis for causal inference. Several participants were excluded from the final analysis due to extreme pre-test scores that could have disproportionately influenced the results. The final sample comprised 57 students, with 30 in the experimental group and 27 in the control group. While the sample size was based on participant availability rather than an a priori power analysis, it is consistent with similar studies in the field and adequate for detecting medium-to-large effects [16,50], a conclusion supported by the post-hoc power analysis presented in Section 4.4.

The experimental group utilized the AI-GSLF, which featured AI-driven real-time feedback and various gamification elements, such as a reward system, character collection, and a leaderboard. The control group used a non-AI, non-gamified speech learning framework (NAI-NGSLF) that provided only core instructional content and self-practice functions. To ensure content equivalency, the instructional materials, key concepts, and speech practice topics were identical across both systems. This study aimed to investigate the effects of the AI-GSLF on learners’ learning behaviors, academic performance, motivation, and immersive experience.

During the intervention, instructional video materials were selected based on the participants’ baseline speaking proficiency. Key concepts and relevant vocabulary from these videos were integrated into both the AI-GSLF and NAI-NGSLF systems. A detailed comparison of the features available to the experimental and control groups is provided in Table 1.

4.2. Experimental procedure

The experiment was conducted over a 30-day period, during which participants could use the assigned learning system at their preferred times and locations. The procedure involved three phases: pre-test, intervention, and post-test.

Pre-Intervention Phase. Before the experiment commenced, all participants completed a speech pre-test to establish baseline public speaking performance. The performances were assessed by two trained raters using a rubric adapted from Roohr, Burkander, and Mao [74]. The rubric evaluated four dimensions: fluency, clarity, intonation, and content organization, each rated on a 5-point scale (1 = very poor to 5 = excellent). The two raters scored all speeches independently, and

Table 1
Comparison of Intervention Content between Experimental and Control Groups.

Dimension	Experimental Group (AI-GSLF)	Control Group (NAI-NGSLF)
Content exposure	Identical video materials, key speech points, and vocabulary	Identical video materials, key speech points, and vocabulary
Feedback	Real-time AI evaluation and adaptive feedback	Self-practice, no automated feedback
Gamification	Rewards, leaderboard, character collection, achievements, role-playing storyline, etc.	None
User interface	Game-like interface and first-person speech scenarios	Simple learning interface, task-based practice
Training scenarios	Workplace report, campus speech, debate, etc.	Same scenarios, without gamified elements and immersion features

inter-rater reliability was high, as measured by a two-way random-effects intraclass correlation coefficient ($ICC(2, k) = 0.87$, 95 % CI [0.82, 0.91]). To prevent practice effects, the topics for the pre-test and post-test differed but were matched for difficulty and task requirements. The complete scoring rubric is provided in Appendix 3 (Table App. 3).

Intervention Phase. Throughout the 30-day intervention, both the AI-GSLF and NAI-NGSLF systems automatically logged user interaction data, including learning duration and usage frequency. The AI-GSLF system additionally recorded the content of all AI-generated feedback. To ensure data integrity, log data were uploaded in real time to the Object Storage Service (OSS) for subsequent processing and analysis.

Post-Intervention Phase. Upon completion of the experiment, all participants undertook a speech post-test, which was evaluated by the same two raters using the same rubric, ensuring consistency in scoring, although the specific test content differed between pre-tests and post-tests to prevent practice effects. A delayed post-test was administered one month after the intervention to assess the long-term retention of skills. Additionally, participants completed the Motivated Strategies for Learning Questionnaire (MSLQ) and an Immersion Experience Scale to measure learning motivation and engagement, respectively (see Appendix 3, Tables App. 4 and 5). Finally, semi-structured interviews were conducted with a subsample of eight students from each group to gather qualitative data on their subjective experiences. The interview transcripts were thematically analyzed to identify key perceptions of the system features.

4.3. Assessment instruments

In addition to performance metrics, this study employed validated psychological instruments to quantitatively assess learners’ affective and cognitive states. The Motivated Strategies for Learning Questionnaire (MSLQ) and the Immersion Experience Scale were selected to provide reliable measures of subjective experiences, including learning motivation and immersion, thereby complementing behavioral and performance data.

4.3.1. The motivated strategies for learning questionnaire (MSLQ)

Learning motivation was assessed using a modified version of the Motivated Strategies for Learning Questionnaire (MSLQ), which was originally developed by Pintrich [52]. While the full MSLQ comprises sections on both motivation and learning strategies, this study utilized only the motivation component to evaluate participants’ goal orientation, value beliefs, self-efficacy, and anxiety related to speech learning.

The adapted instrument consists of 20 items rated on a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*) [75]. It measures four dimensions:

- 1. **Intrinsic Motivation:** Assesses motivation driven by internal factors such as interest, enjoyment, or personal challenge.

- 2. **Extrinsic Motivation:** Evaluates motivation driven by external factors, including rewards, system design, or task requirements.
- 3. **Self-Efficacy:** Measures students' confidence in their ability to improve their speech skills and self-regulate their learning.
- 4. **Anxiety Reduction:** Assesses changes in nervousness and stress related to public speaking tasks.

To ensure suitability for elementary school students, the questionnaire was adapted by simplifying language, avoiding abstract concepts, and incorporating visual elements to maintain engagement [76]. A pilot test confirmed the instrument's reliability. The overall internal consistency was excellent (Cronbach's $\alpha = 0.93$), and the subscales also demonstrated good internal consistency: Intrinsic Motivation ($\alpha = 0.81$), Extrinsic Motivation ($\alpha = 0.80$), Self-Efficacy ($\alpha = 0.82$), and Anxiety Reduction ($\alpha = 0.83$). These results indicate that the adapted instrument was suitable for use in this study.

4.3.2. The immersion experience scale

Participants' immersive experience was measured using the Immersion Experience Scale developed by Cheng et al. [51]. The instrument assesses three core dimensions of immersion:

- 1. **Engagement:** Captures the degree of emotional involvement and active participation while using the system (e.g., "When I use the system, I feel it is fun, and time passes quickly").
- 2. **Engrossment:** Measures the level of focused attention and resistance to distraction during the learning process (e.g., "I am very focused while learning and rarely think about other things").
- 3. **Total Immersion:** Evaluates the extent to which a user feels completely absorbed in the activity, characterized by a sense of presence and reduced awareness of external surroundings (e.g., "When I am learning, I feel completely immersed in the task and forget what is happening around me").

A pilot study confirmed the instrument's high reliability. The overall scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.93$), as did the three subscales: Engagement ($\alpha = 0.80$), Engrossment ($\alpha = 0.81$), and Total Immersion ($\alpha = 0.86$).

4.4. Data analysis

To test the research hypotheses, this study employed a multimodal data collection approach followed by parametric statistical analysis. Data sources included: (1) behavioral logs from the system (e.g., practice frequency, duration); (2) AI-generated diagnostic records (e.g., speech rate, fluency); (3) pre-test and post-test scores; (4) validated psychological questionnaires (MSLQ and Immersion Experience Scale); and (5) semi-structured interviews. This combination of quantitative and qualitative data enabled triangulation, enhancing the validity of the findings. All participants completed the study, and no missing data were recorded for the questionnaires or system logs. The only data cleaning performed was the exclusion of participants with extreme pre-test scores, resulting in a final sample of 57 students (experimental group: $n = 30$; control group: $n = 27$).

The following statistical procedures were conducted. First, descriptive statistics were used to analyze and compare learning behaviors between the two groups. Next, to examine differences in learning motivation and immersion, independent samples *t*-tests were performed. This test was deemed appropriate as the data met the assumptions of normality and homogeneity of variances.

To assess the intervention's effect on speaking performance, an Analysis of Covariance (ANCOVA) was used to compare post-test scores between the groups while controlling for pre-test scores, which were included as a covariate. The necessary assumptions for ANCOVA were checked and met. Furthermore, to explore the relationships between engagement with gamification elements and learning outcomes, Pearson

product-moment correlation analysis was conducted.

Finally, an a priori power analysis indicated that the planned sample size ($N = 57$) was sufficient to detect the expected primary effects with a power greater than 0.8, based on conventional benchmarks [77]. All statistical analyses were performed using SPSS version 27 [78].

5. Experimental results

5.1. Analysis of the difference between the two groups

5.1.1. Analysis of differences in learning behaviors between the two groups

To assess learning engagement, three behavioral metrics were extracted from the system logs: (1) **System Usage Frequency**, defined as the average number of daily sessions in which a participant completed at least one practice activity; (2) **Number of Speeches**, representing the total count of completed speech practice sessions; and (3) **Learning Duration**, the cumulative time in hours a participant spent using the system.

As shown in Table 2, the experimental group demonstrated higher levels of engagement across all three behavioral metrics compared to the control group. Specifically, the experimental group completed a substantially higher number of practice sessions (Median = 103, IQR =) than the control group (Median = 34.5, IQR = [29, 52.25]). Similar disparities were observed for system usage frequency (Experimental: Median = 3.89, IQR = [3.56, 4.21]; Control: Median = 1.29, IQR = [1.12, 1.99]) and total learning duration (Experimental: Median = 57.33, IQR = [49.11, 63.17]; Control: Median = 19.55, IQR = [16.94, 29.75]). Boxplots visualizing the distributions for system usage frequency, number of speeches, and learning duration are presented in Figs. 7–9, respectively.

These quantitative findings were supported by qualitative data from the post-intervention interviews (Table 3). Participants in the experimental group frequently reported that the "AI feedback was helpful" and that they perceived an "improvement in ability." The positive perception of the AI feedback, a feature exclusive to the experimental group, suggests it was a key factor contributing to this increased engagement. Collectively, these behavioral and qualitative data indicate that the AI-GSLF successfully fostered greater student engagement in speech practice activities.

5.1.2. Analysis of differences in learning motivation between the two groups

The results of independent samples *t*-tests, summarized in Table 4, indicate that the experimental group reported significantly higher levels of learning motivation than the control group across all four dimensions (all $p < .001$).

The largest difference was observed for the self-efficacy dimension, where the experimental group ($M = 3.87$, $SD = 0.21$) scored significantly higher than the control group ($M = 3.25$, $SD = 0.25$), $t(55) = 10.14$, $p < .001$, 95 % CI [0.50, 0.72]. Overall, the intervention had a positive effect on all measured aspects of motivation, with small but meaningful effect sizes (Cohen's *d* ranged from 0.23 to 0.33). Furthermore, all findings remained statistically significant after applying a Bonferroni correction for multiple comparisons, underscoring the robustness of these results.

Table 2
Statistics of learning behavior results.

Item	Group	n	Mean	SD
Frequency of system use(times/day)	Experimental	30	3.68	1.17
	Control	27	1.46	0.67
Number of speeches	Experimental	30	101.77	21.70
	Control	27	40.70	18.20
Learning duration(h)	Experimental	30	53.95	17.23
	Control	27	21.64	9.18

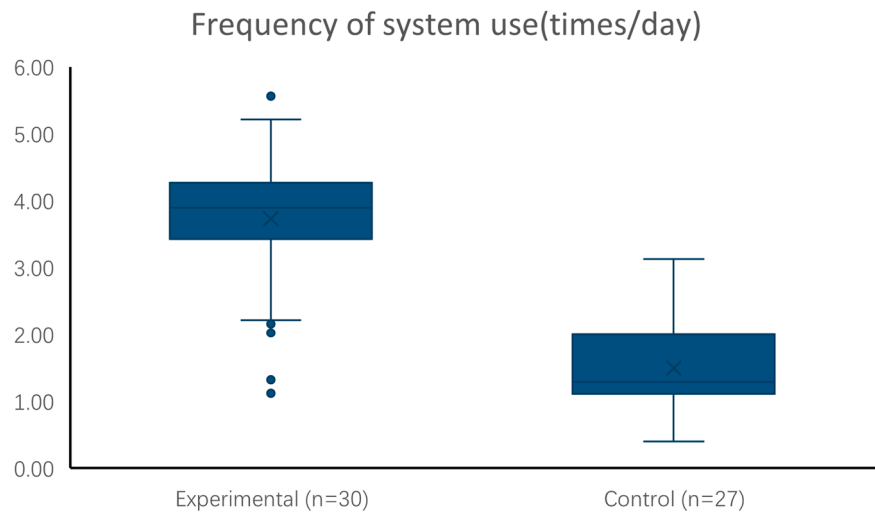


Fig. 7. Boxplot of system usage frequency by group.

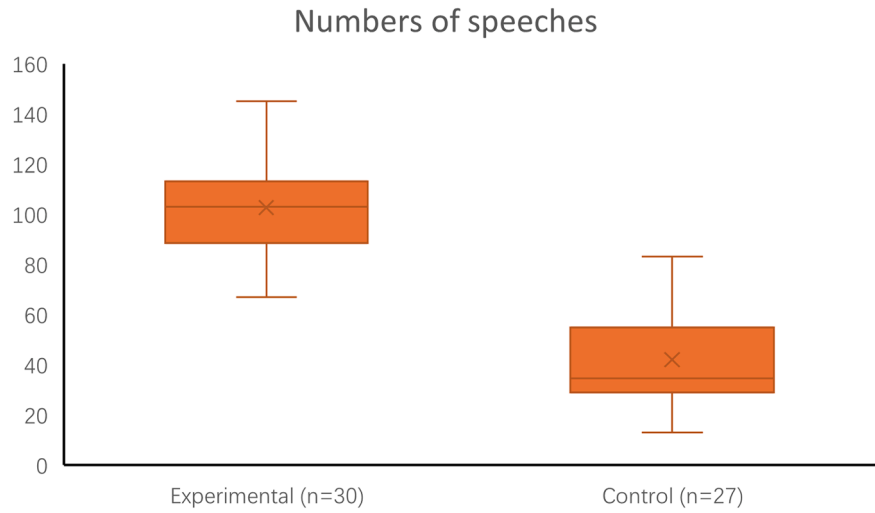


Fig. 8. Boxplot of the number of speeches by group.

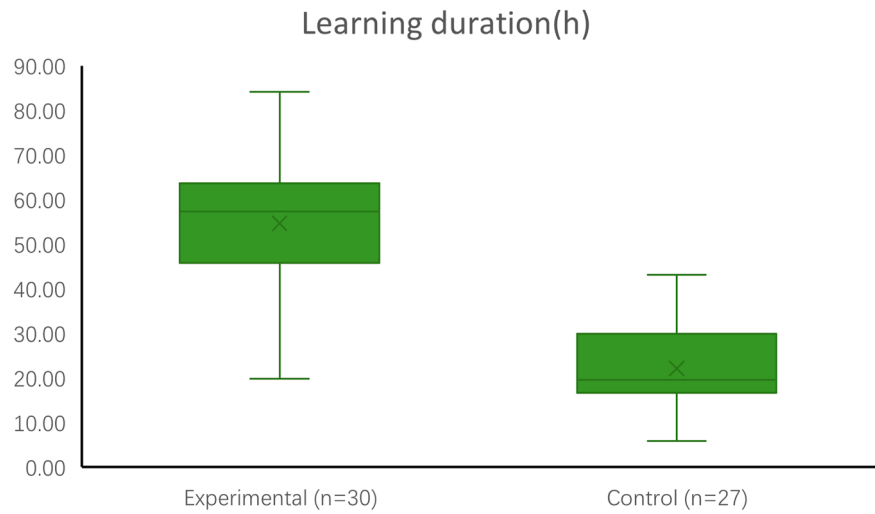


Fig. 9. Boxplot of learning duration by group.

Table 3
Statistics of interview results ($n = 16$).

Main keywords	Mentioned times (experimental group)	Mentioned times (control group)
Interesting	7	4
Innovative learning method	6	3
Challenging content	4	4
AI feedback useful	8	Not applicable
Skill enhancement	8	2

Experimental group $n = 8$; Control group $n = 8$.

5.1.3. Analysis of differences in immersive experience between the two groups

As shown in Table 5, independent samples t -tests revealed that the experimental group reported significantly higher levels of immersive experience than the control group across all three dimensions: engagement, engrossment, and total immersion (all $p < .001$).

Specifically, the experimental group scored higher on engagement ($M = 3.93$ vs. $M = 3.43$), engrossment ($M = 3.85$ vs. $M = 3.36$), and total immersion ($M = 3.92$ vs. $M = 3.44$). The effect sizes for these differences were small (Cohen's $d = 0.25$ to 0.33), with 95 % confidence intervals for the mean differences of $[0.40, 0.64]$, $[0.42, 0.62]$, and $[0.44, 0.61]$ for engagement, engrossment, and total immersion, respectively.

These findings suggest that the AI-GSLF fostered a significantly more immersive learning experience than the control system. The results remained statistically significant after a Bonferroni correction, confirming the robustness of these effects.

5.1.4. Analysis of differences in learning achievement between the two groups

An independent samples t -test was conducted to establish baseline equivalence. The analysis confirmed that there was no significant difference in pre-test scores between the experimental ($M = 63.43$, $SD = 3.17$) and control ($M = 63.42$, $SD = 3.12$) groups, $t(55) = 0.01$, $p = .99$, Cohen's $d < 0.01$, indicating that the groups were comparable prior to the intervention.

Subsequently, an Analysis of Covariance (ANCOVA) was employed to compare post-test performance on speech skills, with pre-test scores included as a covariate. Preliminary assumption checks confirmed the homogeneity of regression slopes, as the interaction between the group

and the covariate was not significant, $F(1, 53) = 0.520$, $p = .474$. A Pearson correlation analysis indicated no significant relationship between pre-test and post-test scores ($r = -0.088$, $p = .515$). However, Levene's test revealed that the assumption of homogeneity of variance was violated for the post-test scores, $F(1, 55) = 13.060$, $p < .001$. As ANCOVA is generally robust to this violation, particularly with similar group sizes, the analysis proceeded. Residuals from the ANCOVA were examined for normality using the Shapiro-Wilk test ($W = 0.936$, $p = .005$), indicating a mild deviation from normality.

As detailed in Table 6, the ANCOVA revealed a significant main effect of the group on post-test scores, $F(1, 54) = 11.669$, $p = .001$, partial $\eta^2 = 0.178$. The experimental group achieved a higher adjusted mean score ($M = 74.13$, 95 % CI $[70.82, 77.44]$) compared to the control group ($M = 66.98$, 95 % CI $[63.55, 70.41]$). The partial eta-squared value indicates a large effect size, suggesting that the AI-GSLF had a meaningful impact on improving students' speech performance.

To assess the long-term effects of the intervention, skill retention was evaluated one month post-experiment. An independent samples t -test was conducted on the achievement decay score (post-test score minus delayed post-test score). As Levene's test indicated unequal variances ($F = 13.203$, $p < .001$), the results for unequal variances are reported (see Table 7). The analysis revealed a significant difference between the groups, $t(43.5) = -2.70$, $p = .015$, with a 95 % CI for the mean difference of $[-2.09, -0.25]$. The experimental group ($M = 1.53$) exhibited significantly less skill decay than the control group ($M = 2.70$). The large effect size (Cohen's $d = 1.64$) indicates that the intervention had a substantial positive effect on learning retention.

5.2. Correlation analysis between learning achievement and learning behaviors in the experimental group

To investigate the relationship between engagement with specific system features and learning outcomes, a Pearson correlation analysis was conducted on the data from the experimental group ($n = 30$). The results, detailed in Table 8, reveal several significant associations between gamified learning behaviors and post-test performance.

The strongest positive correlations were observed between post-test scores and the number of times feedback was viewed ($r = 0.703$, $p < .001$, large) and the total number of speeches delivered ($r = 0.695$, $p < .001$, large). These large effects suggest that frequent practice and active engagement with AI feedback were strongly associated with improved

Table 4
The independent samples t -test results of learning motivation differences between the two groups.

Item	Group	n	Mean	SD	t	p(two-tailed)	Cohen's d
Intrinsic motivation	Experimental	30	3.77	0.28	6.04***	<0.001	0.30
	Control	27	3.30	0.31			
Extrinsic motivation	Experimental	30	3.83	0.22	7.61***	<0.001	0.26
	Control	27	3.30	0.30			
Self-efficacy	Experimental	30	3.87	0.21	10.14***	<0.001	0.23
	Control	27	3.25	0.25			
Anxiety reduction	Experimental	30	3.65	0.34	4.63***	<0.001	0.33
	Control	27	3.25	0.32			

Note. *** $p < 0.001$. Cohen's d effect sizes are interpreted according to Cohen [77]: 0.2 = small, 0.5 = medium, 0.8 = large.

Table 5
The independent samples t -test results of immersive experience differences between the two groups.

Item	Group	n	Mean	SD	t	p(two-tailed)	Cohen's d
Engagement	Experimental	30	3.93	0.12	7.59***	<0.001	0.25
	Control	27	3.43	0.34			
Engrossment	Experimental	30	3.85	0.24	6.48***	<0.001	0.29
	Control	27	3.36	0.33			
Total Immersion	Experimental	30	3.92	0.30	5.43***	<0.001	0.33
	Control	27	3.44	0.37			

*** $p < 0.001$.

Table 6

The ANCOVA results of the post-test score difference between the two groups.

Item	Group	n	Pre-test		Post-test		F	p(two-tailed)	Partial η^2
			Mean	SD	Mean	SD			
Learning achievement	Experimental	30	63.43	3.17	74.13	9.36	11.67**	.001	0.178
	Control	27	63.42	3.12	66.98	5.75			

** $p < 0.01$.**Table 7**The independent samples *t*-test results of the delayed post-test decline scores difference between the two groups.

Item	Group	n	Mean	SD	t	p(two-tailed)	Cohen's d
Decline score	Experimental	30	1.53	0.77	-2.70*	0.015	1.64
	Control	27	2.70	2.25			

* $p < 0.05$.**Table 8**Correlation analysis between learning achievement and learning behaviors in the experimental group ($n = 30$).

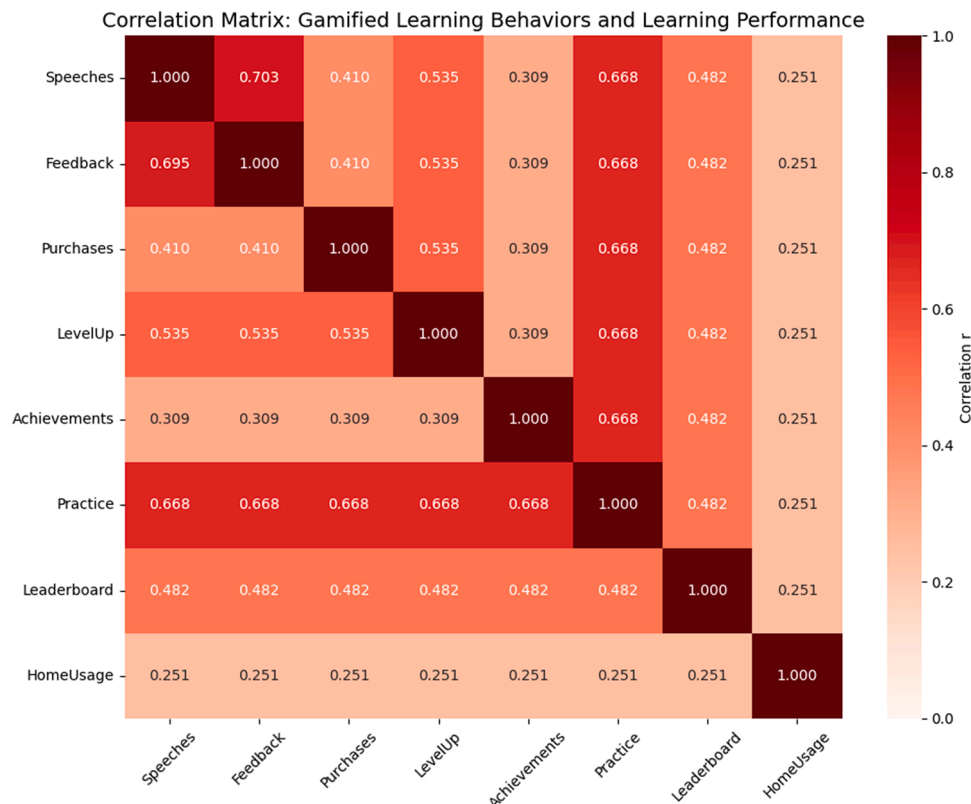
Gamified learning function	Learning achievement progress score	
	r	p(two-tailed)
Number of speeches	0.695***	<0.001
Number of viewing feedback	0.703***	<0.001
Number of purchases/reward draws	0.410*	0.025
Number of level upgrades	0.535**	0.002
Number of achievements collected	0.309	0.097
Number of speech skill practices	0.668***	<0.001
Number of viewing leaderboards	0.482**	0.007
Home system usage time	0.251	0.180

* $p < .05$. ** $p < .01$. *** $p < .001$.

performance. Furthermore, speech skill practice frequency ($r = 0.668$, $p < .001$, medium) and the number of level upgrades ($r = 0.535$, $p = .002$, medium) were also significantly correlated with learning outcomes, indicating that consistent skill practice and progression through the system's difficulty levels were associated with higher scores.

Engagement with competitive and reward-based elements also showed significant, albeit weaker, associations with performance. Specifically, the number of leaderboard views ($r = 0.482$, $p = .007$, medium) and the number of in-game rewards acquired ($r = 0.410$, $p = .025$, medium) were moderately correlated with post-test scores. In contrast, the number of achievements collected ($r = 0.309$, $p = .097$, non-significant) and the time spent in the non-practice "home" area of the system ($r = 0.251$, $p = .180$, non-significant) were not significantly correlated with learning performance.

These relationships are visualized in the correlation heatmap in Fig. 10. The heatmap complements the numerical data in Table 8, using

**Fig. 10.** Correlation heatmap of gamified learning behaviors and learning performance.

a color gradient to represent the strength of the correlations, thereby allowing for a rapid visual assessment of the associations between different user behaviors and learning outcomes. These results indicate that AI feedback has become a pivotal resource actively utilized by learners to drive their own progress. The strong positive correlation between feedback access frequency and achievement highlights the central role of AI feedback as a key pathway for learning enhancement, thereby validating the design rationale of integrating automated, on-demand, and personalized guidance as a core system feature.

6. Discussion

6.1. Summary of key findings

This study investigated the effects of the AI-GSLF on elementary students' learning behaviors, academic performance, motivation, and immersive experience using a randomized controlled design. Analysis of pre-test scores confirmed baseline equivalence between the experimental and control groups, strengthening the conclusion that the observed post-intervention differences are attributable to the AI-GSLF system rather than pre-existing variations between the groups.

The results demonstrate that students who used the AI-GSLF significantly outperformed the control group across all measured outcomes. These findings provide empirical support for the effectiveness of the AI-GSLF in enhancing speech learning and validate the application of gamified, AI-driven approaches in this domain. Furthermore, the study highlights a key practical advantage of the AI-GSLF: its ability to provide automated, personalized feedback. This feature can reduce the reliance on specialized human instructors, thereby potentially lowering the costs and logistical barriers associated with traditional speech training. Consequently, the system offers learners greater flexibility to practice independently at any time, which may contribute to increased practice frequency and overall learning efficiency.

6.2. Interpretation in light of prior research

The findings of this study are consistent with and extend prior research on the application of gamification and AI in educational settings. Consistent with the work of C.-M. Chen et al. [16] and Hew et al. [32], the AI-GSLF was found to significantly increase behavioral engagement. However, whereas Hew et al. [32] observed limited effects on knowledge acquisition, the current study found that the AI-GSLF translated this increased engagement into significant performance gains. This difference may be attributable to the integration of personalized AI feedback. The positive impact of the AI feedback component aligns with findings from Zou et al. [43] and Speeko [12], which underscore the importance of personalized, real-time feedback for developing oral communication skills. This conclusion is further supported by the qualitative interview data from the present study.

In the domains of motivation and immersion, this study addresses a research gap identified by Li and Tsai [17]. The results demonstrate that the AI-GSLF can produce significant improvements in affective and motivational outcomes concurrently with cognitive gains, specifically within a formal speech training context. Finally, the findings on long-term skill retention contrast with those reported by C.-M. Chen et al. [16]. This suggests that the combination of gamification with immediate, AI-driven feedback may promote more durable learning, potentially by facilitating deeper practice and skill automatization.

Despite these positive findings, it is important to acknowledge several limitations and areas for future research. First, the effectiveness of the AI-GSLF is contingent upon technical factors such as device availability, microphone quality, and network stability. These variables could present significant challenges to implementation in resource-constrained environments. Second, while the AI provides timely and personalized guidance on vocal delivery and content, it cannot fully replicate the nuanced, holistic support of a human instructor, who can

interpret complex non-verbal cues and contextualize feedback within broader pedagogical goals. Therefore, future research should investigate models that blend AI-driven feedback with teacher facilitation to create a synergistic learning environment that optimizes both scalability and pedagogical effectiveness.

6.3. Strengths and contributions

A primary strength of this study is its robust theoretical foundation. The design of the AI-GSLF is explicitly grounded in established learning theories—including Deliberate Practice Theory (frequent practice with feedback), Flow Theory (adaptive difficulty), Cognitive Load Theory (micro-unit lessons), and the Feynman Technique (explanation-based tasks)—to provide a coherent pedagogical rationale that extends beyond simple technical validation.

From a practical standpoint, the AI-GSLF offers several distinct advantages over existing general-purpose language learning applications. These include: (1) close alignment with primary school speech education curricula; (2) the provision of actionable, formative feedback that supports real-time improvement and generates teacher-accessible performance logs; and (3) the integration of theory-informed gamification designed to foster deep engagement rather than superficial reward-seeking behaviors.

Finally, the study makes a significant practical contribution through its implementation. The use of the open-source Godot engine enhances the system's reproducibility and scalability, offering a cost-effective model for deployment in resource-constrained educational settings. By leveraging AI to automate personalized feedback, the AI-GSLF reduces the reliance on specialized instructors, thereby increasing the accessibility and potential for equitable distribution of high-quality speech education.

6.4. Limitations of the study

Despite the promising findings, this study has several limitations that warrant consideration and provide directions for future research.

First, the study's sample size ($n = 57$) was relatively small and drawn from a single primary school in China. These factors pose a clear threat to external validity and limit the generalizability of the findings to other demographic, cultural, and educational contexts. Furthermore, the sample size precluded subgroup analyses (e.g., by gender or baseline ability) and resulted in statistical power ranging from 0.52 to 0.72, which may have been insufficient to detect smaller but potentially meaningful effects. Future research should employ larger, more diverse samples, guided by a priori power analyses, to enhance both internal and external validity.

Second, the 30-day intervention period restricts conclusions regarding the long-term sustainability of the observed effects. Although the delayed post-test indicated better skill retention in the experimental group, a longer-term study is necessary to distinguish sustained learning from novelty or Hawthorne effects and to investigate the causal mechanisms underlying skill retention.

Third, the experimental design evaluated the AI-GSLF as an integrated system, making it impossible to isolate the specific contributions of individual components such as gamification or AI feedback. In particular, while the system was designed with reference to the Octalysis framework, the study does not test Octalysis itself. Rather, Octalysis served as a design tool, and the findings demonstrate the effectiveness of the overall package without disentangling the contribution of each individual core drive. Future studies could employ comparative or ablation designs to disentangle the effects of these distinct features. Additionally, the assessment of motivation and immersion relied on self-report measures, which may be susceptible to social desirability bias, particularly in young learners. Integrating behavioral and observational data would provide a more triangulated understanding of these psychological constructs.

Fourth, several contextual factors were not controlled. Participants used the system in varied environments, and the study did not account for the potential influence of factors such as parental support or distractions. Moreover, the minimal role of teachers in this study leaves the potential impact of instructional integration and teacher-AI collaboration unexplored. Future research should consider more standardized implementation protocols and investigate the system's use within blended learning models.

Fifth, the AI feedback mechanism itself has inherent limitations. The system primarily analyzes linguistic and paralinguistic features (e.g., speech rate, clarity) and cannot evaluate non-verbal communication, such as gestures and facial expressions, which are critical to effective public speaking. As a result, the feedback represents only a partial and incomplete assessment of public speaking performance, with improvements captured mainly in verbal delivery. The system's robustness to dialectal variations or background noise was also not systematically evaluated. Future iterations could incorporate multimodal analysis to provide more holistic feedback.

Finally, while the AI-GSLF was developed for Chinese elementary students, its cross-linguistic and cross-cultural applicability requires empirical validation. Although the system's open-source nature facilitates adaptation, the reliance on commercial cloud services for AI processing and data storage introduces potential operational costs that must be considered when evaluating its long-term scalability and cost-effectiveness.

6.5. Educational implications

The findings of this study offer several practical implications for educators, instructional designers, and policymakers:

- **Evolving Role of the Teacher:** Rather than being replaced by technology, the teacher's role can evolve to that of a facilitator. In this capacity, educators can help students interpret AI-generated feedback, scaffold practice activities, and connect gamified achievements to broader, meaningful learning objectives.
- **Curriculum Integration:** To maximize its educational impact, the AI-GSLF should be integrated into existing curricula, such as oral language modules or project-based learning activities, rather than being used as a standalone supplement. This alignment ensures that the skills developed are directly relevant to established learning standards.
- **Centrality of Feedback in Learning Design:** The strong association between feedback engagement and achievement emphasizes that AI-driven feedback should be treated as a core element of learning design rather than an auxiliary feature. Ensuring that feedback is timely, accessible, and actionable can empower learners to self-regulate their practice and accelerate improvement.
- **Accessibility in Resource-Limited Settings:** For deployment in educational contexts with limited internet connectivity or device access, the development of simplified, mobile-first, or offline-capable versions of the system could significantly enhance its viability and accessibility.
- **Teacher Professional Development:** Effective implementation requires professional development programs that equip teachers with the skills to critically evaluate and utilize automated feedback. Such training should focus on strategies for balancing technology-driven engagement with deep pedagogical objectives.
- **Domain-Specific Design Framework:** The system's design, which concurrently targets foundational speech skills and specific academic content, provides a valuable model for the development of other domain-specific educational tools. This dual-alignment approach is designed to promote the transfer of skills to authentic classroom contexts.

7. Conclusions and future works

This study investigated the effectiveness of an AI-driven gamified speech learning framework (AI-GSLF) for improving the public speaking skills of elementary school students. By conducting a randomized controlled trial comparing the AI-GSLF with a non-gamified, non-AI control version (NAI-NGSLF), the study provides empirical evidence addressing its core research questions.

Q1: Can the AI-GSLF improve learners' engagement and practice frequency at the behavioral level?

Participants using the AI-GSLF demonstrated significantly higher engagement than the control group, as reflected in system usage frequency, number of practice sessions, and total learning duration. This indicates that the gamified elements effectively fostered learner initiative, sustained practice, and autonomous engagement.

Q2: Can the AI-GSLF enhance learners' learning motivation and sense of immersion?

Regarding affective and cognitive outcomes, learners in the experimental group reported significantly higher motivation and stronger immersion, particularly in self-efficacy and focus. These findings suggest that the AI-GSLF's gamification design successfully stimulates interest, promotes engagement, and supports a more immersive learning experience.

Q3: Can the AI-GSLF more effectively enhance learners' key sub-skills in public speaking, including Fluency, Clarity, Intonation, and Content Organization?

In terms of skill development, the experimental group achieved significantly greater improvements in speech fluency, clarity, intonation, and content organization. Additionally, performance in the delayed post-test remained relatively strong, indicating that the AI-GSLF supports not only short-term skill acquisition but also longer-term retention. Notably, the system's use of real-time AI-driven feedback, combined with the Octalysis gamification framework, provides learners with personalized, scaffolded guidance that addresses the common challenges of delayed feedback and limited individualized instruction in traditional speech education.

7.1. Key contributions

1. **Empirical Validation:** Demonstrates that combining gamification and AI-driven feedback can enhance speech competence in elementary students.
2. **Practical Solution:** Introduces a low-cost, real-time speech rate detection algorithm that enables continuous and autonomous practice.
3. **Expanded Domain of Gamified Learning:** Extends the application of gamified educational approaches to public speaking, offering a model for integrating skill-specific training with academic content.
4. **Central Role of Feedback:** Identifies AI-driven, on-demand feedback as a central pathway for learning gains, with strong correlations showing that learners who engaged more with feedback achieved higher performance. This highlights feedback not merely as a feature but as a core success factor in system design.

7.2. Strengths of the study

This study also has several notable strengths. First, it integrates a comprehensive gamification framework (Octalysis) with an AI-powered feedback mechanism, representing an innovative combination that has rarely been applied in formal speech education. Second, it develops and validates a low-cost, real-time speech rate detection algorithm, making scalable classroom deployment feasible without expensive hardware. Third, the study employs a rigorous true experimental design, including random assignment and baseline equivalence checks, which strengthens the internal validity of the findings. Finally, it focuses on primary school students—a young and under-researched population in speech

education—thereby extending the applicability of AI-driven learning systems to a context of both practical and theoretical importance.

7.3. Limitations and future research

Despite these positive findings, several limitations should be acknowledged:

- **Sample and Generalizability:** The study involved 57 students from a single primary school in China, limiting geographic, cultural, and socioeconomic generalizability.
- **Intervention Duration:** The 30-day period constrains conclusions regarding long-term retention; longitudinal studies are needed.
- **Environmental Variability:** Home learning conditions, device differences, and voluntary practice times were not fully controlled, potentially introducing confounds.
- **Component-Specific Effects:** The integrated system design precludes isolating the effects of individual features such as AI feedback or gamification.
- **AI Feedback Limitations:** The system analyzed verbal and paralinguistic features but could not capture non-verbal communication (gestures, facial expressions) or simulate realistic audiences.

Future research should aim to expand participant diversity, extend longitudinal follow-ups, incorporate multimodal feedback (audio + visual), and examine component-specific effects to rigorously evaluate scalability, effectiveness, and educational relevance.

This study confirms that AI-GSLF significantly improves elementary students' public speaking skills, engagement, motivation, and immersive experience. By integrating gamification with AI-driven real-time feedback, the framework provides a personalized, scalable, and effective approach to speech learning, offering valuable insights for the design of flexible, cost-effective educational technologies.

Data availability

The datasets generated and analyzed during the current study are available in two forms. The actual participant data cannot be shared due to privacy constraints involving minors. However, anonymized/sample datasets generated during software testing, illustrating the data structure and key variables, are publicly available at: [<https://github.com/JackeHsu/Strongest-Speech-Streamer>]. In addition, upon acceptance of the manuscript, the analysis scripts and workflow documentation will be deposited in a recognized repository (e.g., Zenodo or Figshare), ensuring transparency and reproducibility.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, Gemini 2.5 Pro was used in some sections of the manuscript to improve readability and grammatical accuracy. After using this tool, the author(s) reviewed and edited the content as needed, and take full responsibility for the final version of the manuscript.

CRedit authorship contribution statement

Xing Sun: Writing – review & editing, Supervision, Methodology, Conceptualization. **Zi-Xiang Xu:** Writing – review & editing, Writing – original draft, Validation. **Ling-Chen Meng:** Software, Investigation, Data curation. **Ding-Nan Shi:** Validation, Software, Resources.

Declaration of competing interest

The author declares that there is no conflict of interest regarding the publication of this paper. No financial support, sponsorship, or personal

relationships that could influence the research outcomes were involved.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.caeo.2025.100312](https://doi.org/10.1016/j.caeo.2025.100312).

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